

Wireshark

- **Graphical user interface (GUI):** Wireshark provides a visual interface with packet-list, packet-details, and packet-bytes panes, making packet analysis easier to navigate.
- **Advanced analysis and filtering:** It provides both capture filters and display filters, extensive protocol dissection, packet colorization, searching, and statistics.
- **Typical use:** Security analysts commonly use Wireshark for detailed packet inspection, troubleshooting network problems, investigating security issues, and analyzing saved packet captures.
- **Consideration:** Its graphical interface and broad analysis features generally require more system resources than a lightweight command-line tool.

Similarities

- Both are network protocol analyzers used to capture and examine network traffic.
- Both can filter network traffic so analysts can focus on packets that match specific protocols, addresses, ports, or other criteria. Wireshark supports capture and display filters, while tcpdump uses libpcap filter expressions.
- Both are open-source and freely available. Wireshark is released under the GNU GPL, while tcpdump is distributed under a BSD license.
- Both can work with saved packet captures, allowing traffic to be examined after it has been collected.

tcpdump

- **Command-line interface (CLI):** tcpdump is a command-line packet analyzer designed to capture and display network traffic from a terminal.
- **Lightweight and scriptable:** Its command-line design makes it well suited for remote servers, automated scripts, and systems without a graphical desktop.
- **Filtering and capture:** tcpdump uses Boolean packet-filter expressions based on the libpcap filtering language and can save captured traffic to a file or read previously saved captures.
- **Typical use:** Analysts often use tcpdump for quick packet capture, troubleshooting, monitoring, and collecting traffic for later analysis in another tool.